Stats English

August 4, 2026
2:35 PM

Mode

The **mode** is the value that appears most frequently. If two non-adjacent values appear more frequently than the others, the distribution is said to be **bimodal**, even if the frequencies of the two modes are not identical.
The mode is used when the **most likely or most frequent value** is of interest. For example, a clothing store manager planning to stock a new dress design will first order dresses in the **modal size**, since those are the sizes she is most likely to sell.
List: 3, 3, 5, 6, 3, 4, 6, 8, 9
Mode = 3

Median

The **median** is the value of the middle element when all the data values are arranged in ascending order. If there are **n** data values, the median is the value of the **(n + 1) / 2**th element. If **n** is odd, there is a single middle value, which is the median.
The median is particularly useful when some values are missing. For example, if 50 people take part in a marathon, the median lies halfway between the **25th and 26th** finishing positions. If some participants do not finish the race, it would be impossible to calculate the average finishing time, but the median is still easy to determine.
When calculating average salaries, the median is often a more appropriate measure than the mean because a few people earning very high salaries can have a large effect on the mean, but not on the median.

Variance and Standard Deviation

Variance and **standard deviation** both measure the **spread** or **variability** of your data around the mean. They indicate how **consistent** your data values are.

Variance = *average of squared differences from the mean.*

- **High variance** → data points are spread out.
- **Low variance** → data points are close to the mean.

The **standard deviation** is just the **square root of the variance**.

$$\text{standard deviation} = \sqrt{\frac{\sum x^2 f}{n} - \bar{x}^2}$$

Find the mean and standard deviation of this set of data
11, 13, 14, 16, 18.

median 17

3, 3, 3, 3, 3

$$Mean = \frac{11+13+14+16+18}{5} = 14.4$$

$$median = \left(\frac{n+1}{2}\right) = \frac{5+1}{2} = \cancel{6}\ 3^{rd}\ position$$

Lower quartile $= 25\% = \frac{n+1}{4} = \frac{6}{4} = 1.5$ pos $= 12$

UQ $= 75\% =$

$$\frac{3}{4} \times (n+1) = \frac{3}{4} \times 6 \quad 4.5 = 17$$

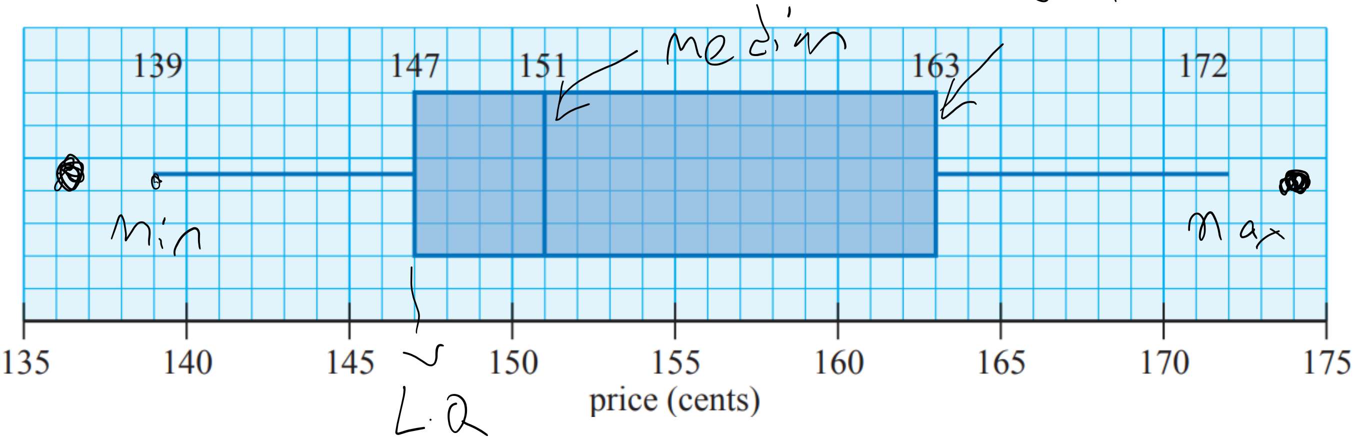
$$var = \frac{x^2}{n} - \bar{x}^2 = \frac{11^2+13^2+14^2+16^2+18^2}{5} - 14.4^2$$

var $= 5.84$

S.D $= \sqrt{var} = 2.41$

Box-and-whisker plots (boxplots)

The three quartiles and the two extreme values of a data set may be illustrated in a box-and-whisker plot.



Data which are more than 1.5 × IQR beyond the lower or upper quartiles are regarded as outliers.

Skewness: Trent Psychology Modules

I.QR = U.Q − L.Q

Question:

Question

The following data represent the annual amounts of money spent on clothing, rounded to the nearest **$10**, by **27 people**.

i) Find the **median** and the **interquartile range (IQR)** of the data.

ii) List the **outliers** in the data.

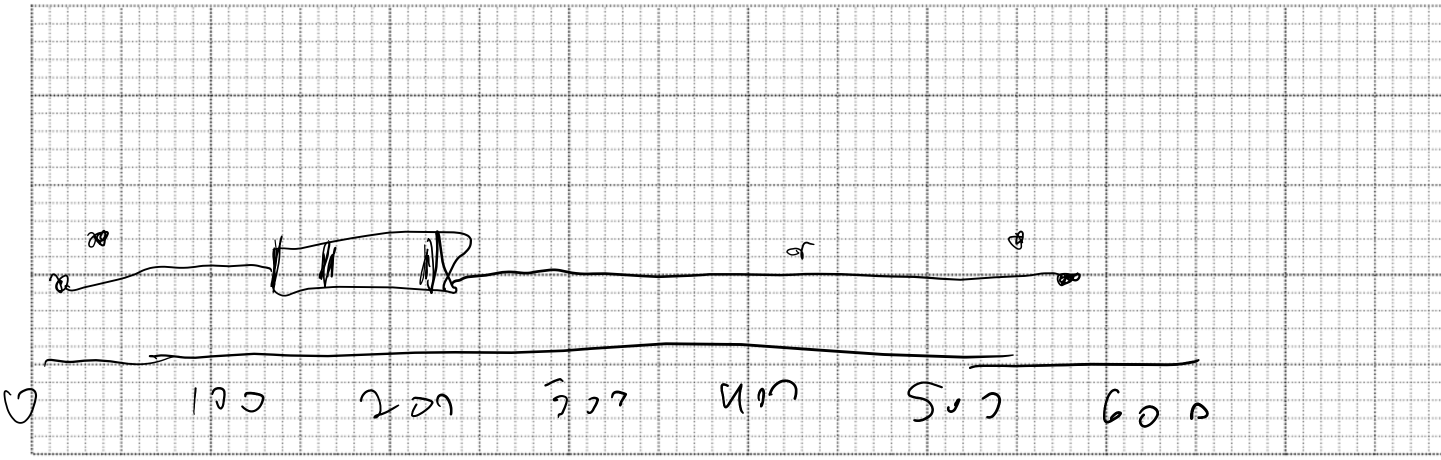
10	40	60	80	100	130	140	140	140
150	150	150	160	160	160	160	170	180
180	200	210	250	270	280	310	450	570

$\text{Median} = \frac{1}{2}(n+1) = \frac{1}{2}(28) = 14^{th}\ \text{pos} = 160$

$U.Q = \frac{3}{4}(n+1) = \frac{3}{\cancel{4}}(\cancel{28}^{7}) = 21^{st}\ \text{pos} = 210$

$L.Q = \frac{1}{4}(n+1) = \frac{1}{4} \times 28 = 7th = 140$

$I.QR = U.Q - L.Q = 210 - 140 = 70$



An **outlier** is defined as any data value that is **more than 1.5 times the interquartile range (IQR) above the upper quartile (Q3)** or **more than 1.5 times the interquartile range (IQR) below the lower quartile**

$I.QR = 70$

$1.5 \times 70 = 105$

$U.Q = 210$

outliers => $> (210 + 105) => > 315$

$LQ = 140$

$140 - 105 = 35$

< 35

Linear Regression

Predict one variable (*Y*) using **one** predictor (*X*).
It assumes a **straight-line relationship** between them.
Y=a+bXY = a + bXY=a+bX

X = num of rooms

Y = house price

Rationale

Linear regression is introduced to conveniently present a well-known training algorithm, **gradient descent**. Additionally, it serves as a foundation for introducing **logistic regression**–a classification algorithm—which further facilitates discussions on **artificial neural networks**.

- Linear Regression
  - Gradient Descent
  - Logistic Regression
    - Neural Networks

Supervised Learning - Regression

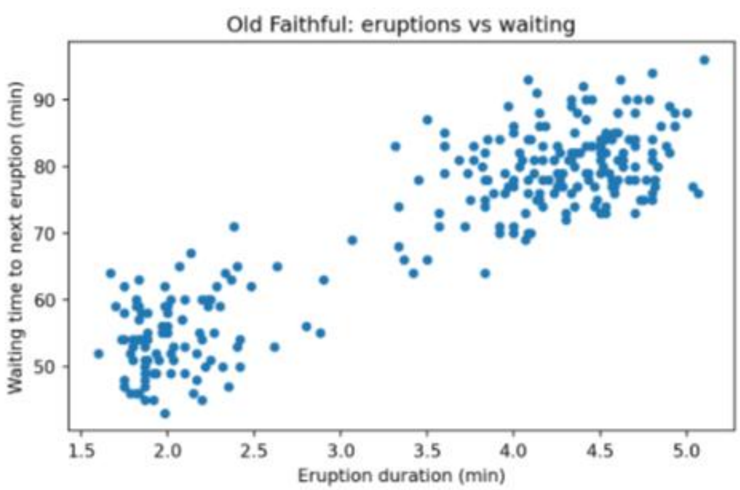
- The **training data** is a collection of **labelled** examples.
  - $\{(x_i, y_i)\}_{i=1}^N$
    - Each x_i is a **feature vector** with D dimensions.
    - $x_i^{(j)}$ is the value of the **feature** j of the example i, for $j \in 1 \ldots D$ and $i \in 1 \ldots N$.
  - The **label** y_i is a **real number**.
- **Problem**: Given the data set as input, create a **model** that can be used to predict the value of y for an unseen x.

Notebook for linear regression:
https://colab.research.google.com/drive/157EL_A18skbTGaFvnvPeZORPftpPu5Uj?usp=sharing

Old Faithful Eruptions

(272, 2)

	eruptions	waiting
0	3.600	79
1	1.800	54
2	3.333	74
3	2.283	62
4	4.533	85
5	2.883	55



Linear Regression

A **linear model** assumes that the value of the **label**, $\hat{y_i}$, can be expressed as a **linear combination** of the feature values, $x_i^{(j)}$:

$$\hat{y_i} = \theta_0 + \theta_1 x_i^{(1)} + \theta_2 x_i^{(2)} + \ldots + \theta_D x_i^{(D)}$$

Here, θ_j is the jth parameter of the (linear) **model**, with θ_0 being the **bias** term/parameter, and $\theta_1 \ldots \theta_D$ being the **feature weights**.

Definition

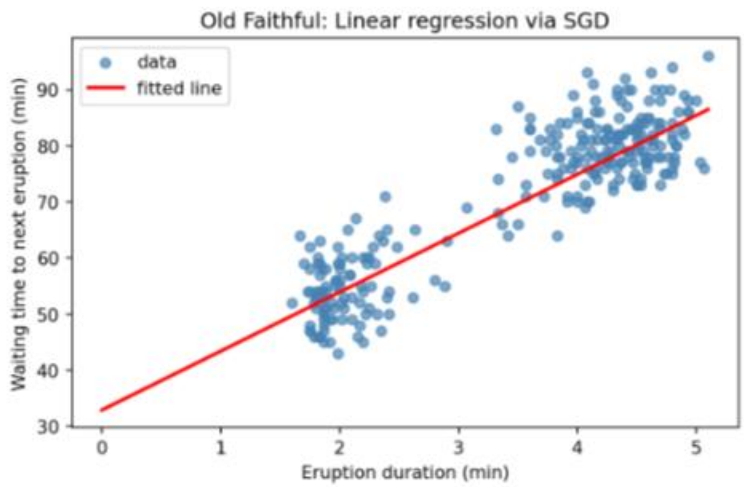
Problem: find values for all the model parameters so that the model **"best fits"** the training data.

- The **Root Mean Square Error** is a common performance measure for regression problems.

$$\sqrt{\frac{1}{N}\sum_{1}^{N}[h(x_i) - y_i]^2}$$

Visualization

▶ Code



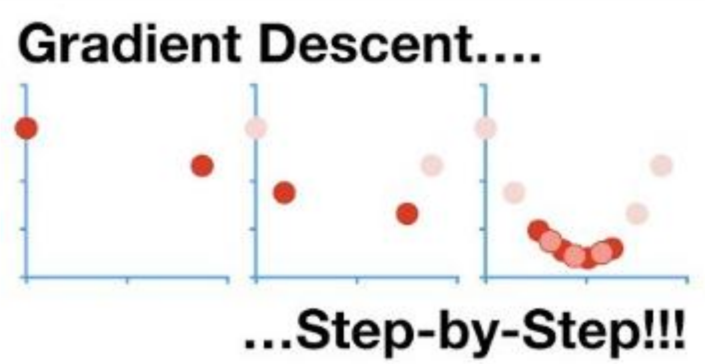
Optimization

Until some termination criteria is met[1]:

- **Evaluate** the loss function, comparing $h(x_i)$ to y_i.
- **Make small changes to the parameters,** in a way that reduces the value of the loss function.

1. E.g. the value of the **loss function no longer decreases** or the **maximum number of iterations.**

[Gradient Descent, Step-by-Step]



Multiple Linear Regression

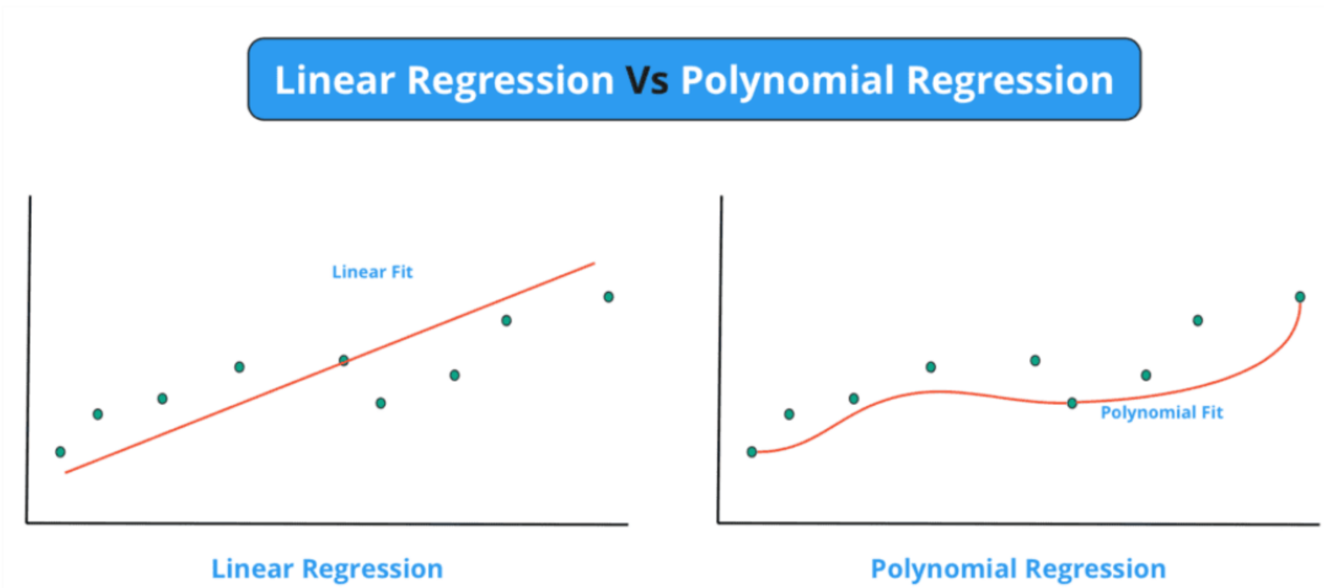
use **more than one X variable** to predict Y.
Y=a+b1X1+b2X2+⋯+bnXnY
For example:

Predicting **house price (Y)** using:

- **Size (X_1)**
- **Number of rooms (X_2)**
- **Age (X_3)**

Polynomial Regression

Used when the relationship between X and Y is **not linear** (the data curves).
Y=a+b1X+b2X^2+b3^X3



[Why Linear regression for Machine Learning?]

KNN - Learning

- *Lazy learning*: no explicit training
- The “model” is the dataset

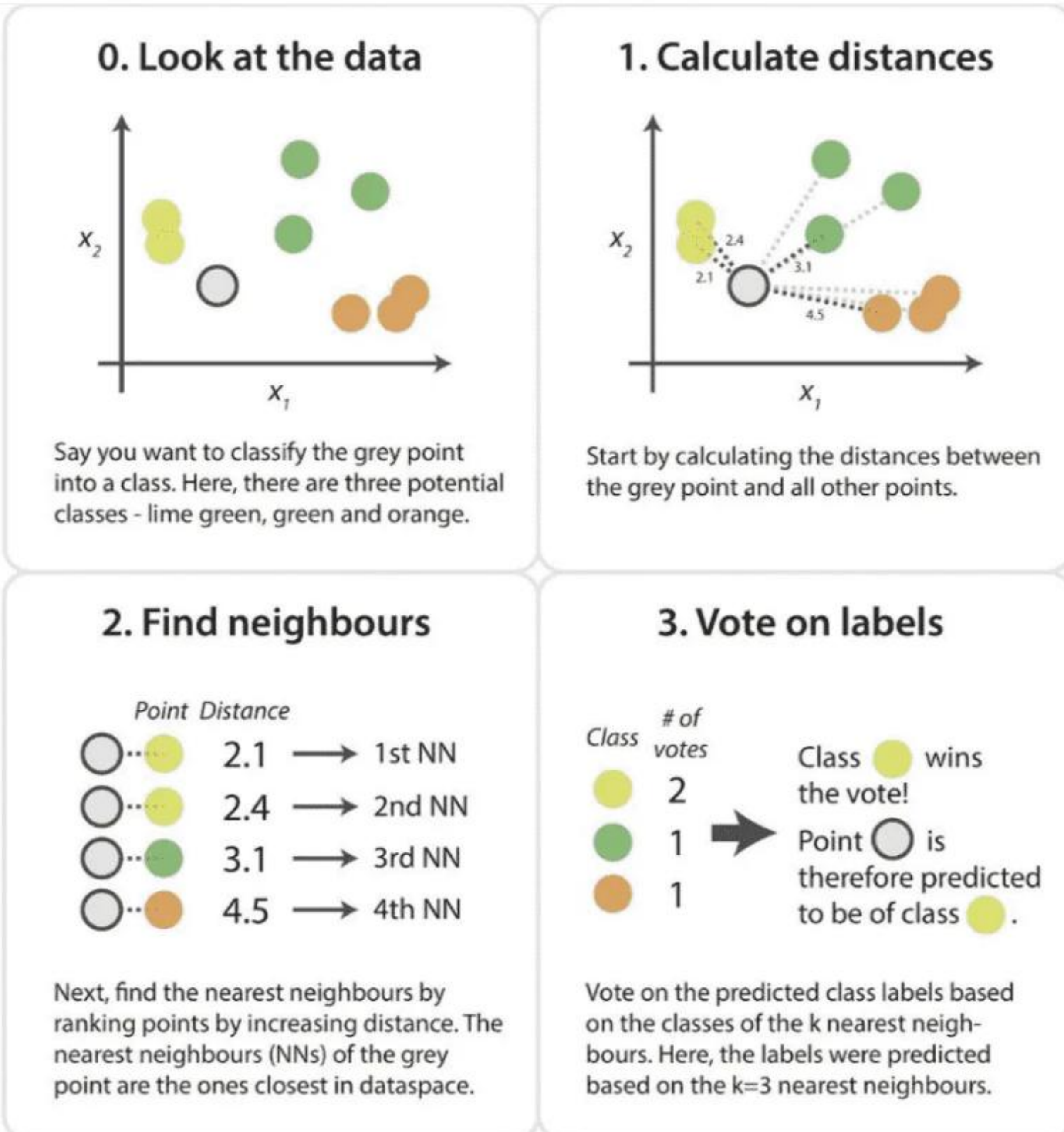
KNN - Inference

- Classify/regress based on the labels/values of the k closest examples in feature space

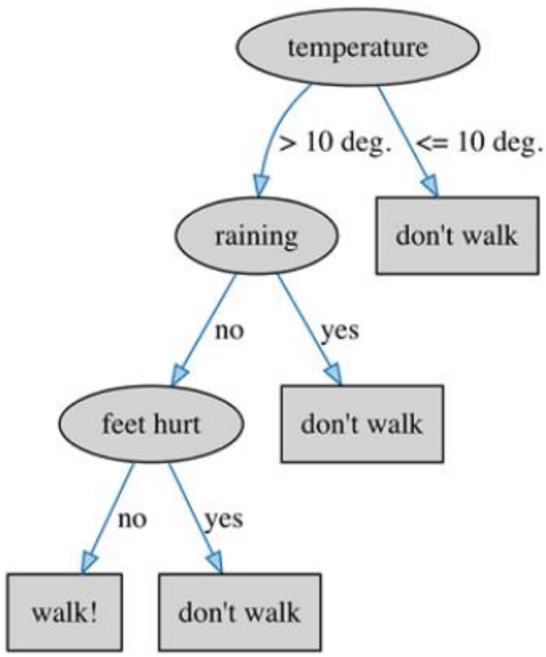
Formal Definition

Given **dataset** $\{(x_i, y_i)\}_1^N$ and an **unseen example** x:

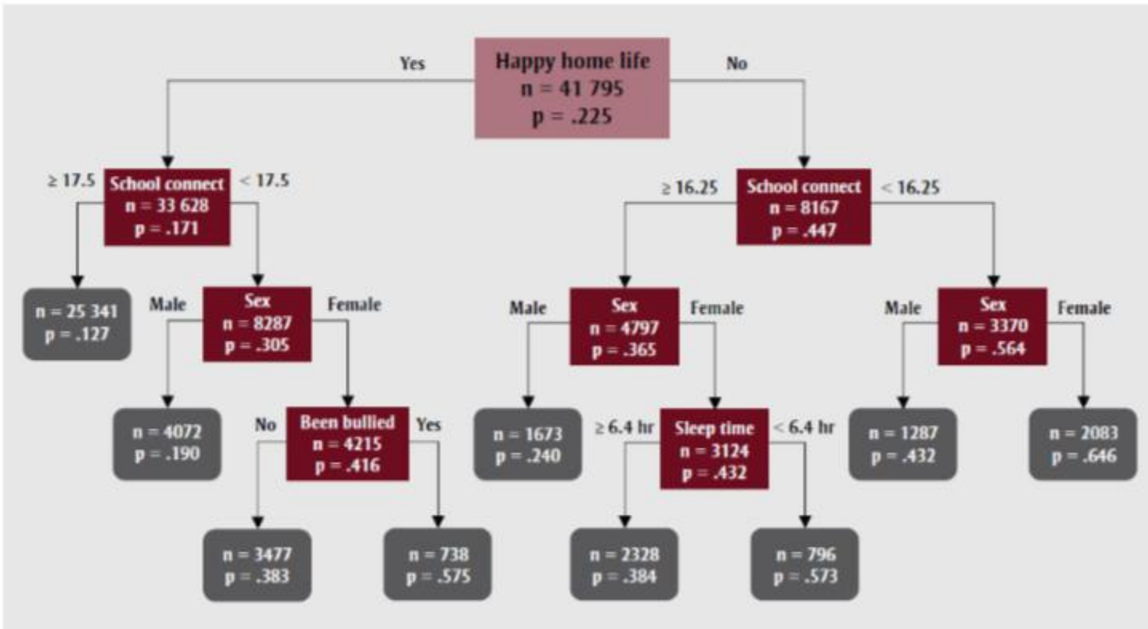
- Compute **distances** $d(x, x_i)$
- Select the k **smallest distances**
- Classification: **majority vote** (possibly weighted by $1/d$)
- Regression: **average** (possibly weighted)


0. Look at the data
x_2
x_1
Say you want to classify the grey point into a class. Here, there are three potential classes - lime green, green and orange.
1. Calculate distances
x_2
2.4
2.1
3.1
4.5
x_1
Start by calculating the distances between the grey point and all other points.
2. Find neighbours
Point Distance
2.1 → 1st NN
2.4 → 2nd NN
3.1 → 3rd NN
4.5 → 4th NN
Next, find the nearest neighbours by ranking points by increasing distance. The nearest neighbours (NNs) of the grey point are the ones closest in dataspace.
3. Vote on labels
Class
# of votes
2
1
1
Class wins the vote!
Point is therefore predicted to be of class .
Vote on the predicted class labels based on the classes of the k nearest neighbours. Here, the labels were predicted based on the k=3 nearest neighbours.

Decision Tree



Interpretable



What is a Decision Tree?

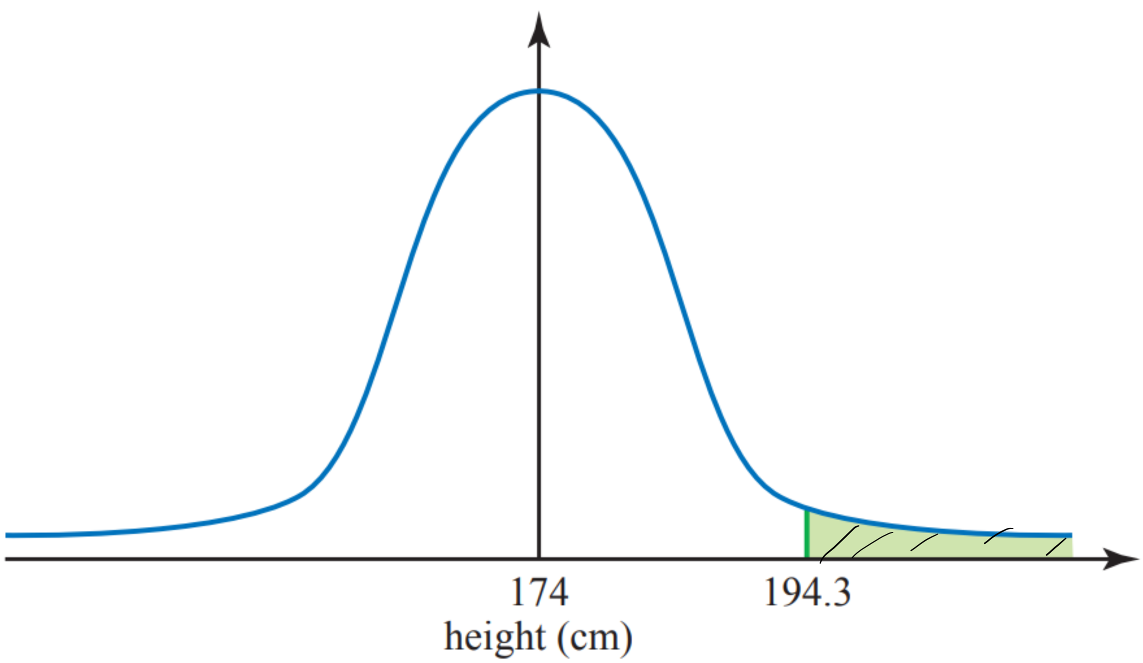
- A decision tree is a **hierarchical structure** represented as a directed acyclic graph, used for **classification** and **regression** tasks.
- Each **internal node** performs a binary test on a particular feature (j), such as evaluating whether the number of connections at a school surpasses a specified threshold.

The tree's **structure** is **inferred** (**learnt**) from the **training data**.

Normal Distribution

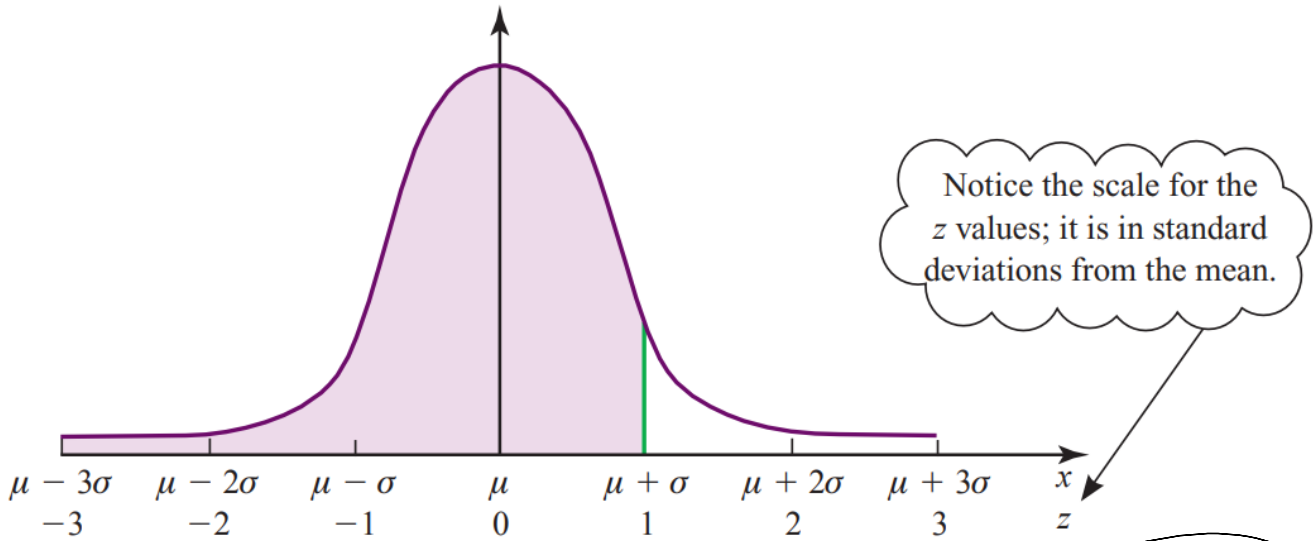
Example: Height is a **continuous variable** and not a discrete variable.
If you measure height with enough precision, it can take **any possible value**.
This means that it is not really meaningful to ask:
"What is the probability that a randomly selected person is exactly 194.3 cm tall?"
The answer is **zero**.
However, you can ask questions such as:
"What is the probability that a randomly selected person is between 194.25 cm and 194.35 cm tall?"
and
"What is the probability that a randomly selected person is at least 194.3 cm tall?"
When the variable is continuous, you are interested in a **range of values** rather than a single value.

Like many other natural variables, the height of adult men can be modeled using a **normal distribution**, as shown in the figure.
You will notice that it has a characteristic **bell-shaped curve** and is **symmetrical around its center**.
The curve is **continuous** because height is a continuous variable.
In the figure, the **area under the curve represents probability**, so the shaded area to the right of **194.3 cm** represents the probability that a randomly selected adult man is **taller than 194.3 cm**.



The function **Φ(z)** gives the **area under the normal distribution curve to the left of the value z**, which is the shaded area shown in the figure. (This is called the **cumulative distribution function**.)

The **total area under the curve is equal to 1**, and the area represented by **$\Phi(z)$** corresponds to the **probability of obtaining a value less than z**.



The shaded area is $\Phi(1)$

=Standardisation

10, 50, 1000,

Standardisation is used to put variables on the same scale so that one variable does not unfairly dominate another because of its units or size.

Standardisation means **transforming all our data so that the mean becomes 0 and the standard deviation becomes 1**.
If the variable **X** has a mean **μ (mu)** and a standard deviation **σ (sigma)**, then a particular value **x** of **X** is transformed into a **z-score** using the equation:

$$z = \frac{x - \overbrace{\mu}^{\to \text{mean}}}{\underbrace{\sigma}_{\to \text{s.d}}}.$$

The **z-score (z)** is a particular value of the variable **Z**, which has a **mean of 0** and a **standard deviation of 1**. It is the **standardized form of the normal distribution**.

	Actual distribution, X	Standardised distribution, Z
Mean	μ	0
Standard deviation	σ	1
Particular value	x	$z = \frac{x - \mu}{\sigma}$

Question:

Assume that the distribution of adult male heights is **normal**, with a **mean of 174 cm** and a **standard deviation of 7 cm**. Calculate the probability that a randomly selected adult man is:

(i) Less than 185 cm

You must standardize first.

Given:

- **X = 185 cm**
- **Mean (μ) = 174 cm**
- **Standard deviation (σ) = 7 cm**

The formula is:

$$z = \frac{x - \mu}{\sigma}$$

$$z = \frac{185 - 174}{7} = 1.5714$$

Using **R**, a programming language for statistics:

$$\text{pnorm}(1.57) = 0.94$$

Therefore, the probability that an adult man is **less than 185 cm** is:

$$\boxed{0.94 \text{ or } 94\%}$$

(ii) More than 185 cm

Since the probability of being **less than 185 cm** is **0.94**, and the total area under the normal curve is **1**:

$$P(X > 185) = 1 - 0.94$$

$$= 0.06$$

Therefore, the probability that an adult man is **taller than 185 cm** is:

$$\boxed{0.06 \text{ or } 6\%}$$

(iii) Between 180 cm and 185 cm

P(185) − P(180)

=For **X = 180 cm**:

- Mean (μ) = 174 cm
- Standard deviation (σ) = 7 cm

$$z = \frac{180 - 174}{7} = 0.8571$$

For **X = 185 cm**:

- Mean (μ) = 174 cm
- Standard deviation (σ) = 7 cm

$$z = \frac{185 - 174}{7} = 1.5714$$

The probability between these two values is:

$$P(180 < X < 185) = \text{pnorm}(1.57) - \text{pnorm}(0.8571)$$

$$= 0.1374873$$

Therefore, the probability that an adult man's height is **between 180 cm and 185 cm** is:

$$\boxed{0.1375 \text{ or } 13.75\%}$$

Sampling

A **sample** provides a set of data values from a random variable, taken from all possible values in the **population** (the parent population). The parent population can be **finite**, such as the set of all professional football players, or **infinite**, such as the points where a dart can land on a target.
The elements available to be sampled are called the **sampling frame**. It can be, for example, a list of sheep in a flock, a map marked with a grid, or an electoral register. In many situations, no sampling frame exists and it is not possible to create one, for example, for North Atlantic cod.
The proportion of available elements that are actually sampled is called the **sampling fraction**.

There are many ways to interpret data. First, you must examine how the sample data is collected and what measures you can take to ensure its quality.
An estimate of a parameter derived from sample data will generally be different from its true value. The difference is called **sampling error**.
To reduce sampling error, you want your sample to be as representative of the parent population as possible. However, this can be easier said than done.

Are the data relevant?

A common mistake is to replace what you actually need to measure with something else simply because the data is easier to obtain.
You must ensure that your data is **relevant**, meaning that it provides values that exactly correspond to what you want to measure.

Are the data likely to be biased?

Bias is a systematic error.
For example, if you wanted to estimate the average time it takes young women to run 100 metres, but you measured the times of members of a hockey team, your result would be biased.
The hockey players would probably be fitter and more athletic than the average young woman, so your estimated time would be too low.
Therefore, bias must be avoided when selecting a sample.

Does the collection method distort the data?
The data collection process should not interfere with the data.
For example, when designing a questionnaire, it is very easy to phrase questions in a way that influences the answers.
Questions such as:

- "Are you a law-abiding citizen?"
- "Do you think your driving is above average?"

naturally encourage people to answer **"Yes."**
In the case of a survey about voting intentions, another problem arises: voting is secret, and asking people who they plan to vote for is asking them to reveal private information.
Some people may find this intrusive and deliberately give a false answer.
People often tend to give the answer they think the researcher wants to hear.

Is the right person collecting the data?
Bias can also be introduced by the choice of the person collecting the data.
For example, if a school administration wants to estimate the proportion of students who smoke (which violates school rules), and each teacher is asked to question five students about whether they smoke, some smokers will almost certainly answer **"No"** to their teacher because they fear punishment.
However, they might answer **"Yes"** to another person.

Is the sample large enough?
The sample must be large enough for the results to have **statistical significance**.
In the example mentioned, the goal was to compare support for four candidates, and a sample of 30 people is completely insufficient.
For opinion polls, a sample size of around **1,000 people** is common.
The required sample size depends on the desired accuracy of the results.
For example, in election polls, a much larger sample is needed if you want a **margin of error of 1%** compared with a **margin of error of 5%**.

Sampling Methods
Stratified Sampling
dividing your population into **groups (called strata)** that share something in common like age, gender, or income.
Then, you take a **sample from each group**.

Example:
You want to survey 120 students about the cafeteria food.
The school has:

- 200 boys
- 400 girls

To ensure both groups are fairly represented, you divide the students into two strata (boys and girls), then choose proportional samples, for example:

- 40 boys

- 80 girls

Purpose: To make sure each important subgroup is represented in the sample.

Cluster Sampling

Cluster sampling divide the population into **clusters (mini-groups)** that each represent the whole population, and then **randomly select a few clusters** to study **completely**.

A country has 1000 schools, and you want to study students’ reading habits.
Instead of choosing students from all schools , you randomly pick 10 schools (clusters) and survey **all the students** in those 10 schools.
It saves time and money when the population is too large or spread out.

Example of Cluster Sampling (Using Neighborhoods and Diversity)
Suppose a city wants to study the health habits of its residents.
The city is very large and contains many neighborhoods where people from different backgrounds live.
Step 1: Create the Clusters
Instead of separating people directly by race, the city divides the population into geographic clusters, such as:

- Neighborhood A
- Neighborhood B
- Neighborhood C
- ...up to 200 neighborhoods

Each neighborhood contains a mix of different backgrounds, so each neighborhood represents the overall population on a smaller scale.

Step 2: Randomly Select Clusters
Instead of surveying people across the entire city, the researchers randomly select:

- 10 neighborhoods out of the 200.

Step 3: Study Everyone in the Selected Clusters
Next, they survey all the residents living in the 10 selected neighborhoods.

Why This Is Cluster Sampling
Because:

- Entire groups (the neighborhoods) are selected.
- Each group already contains a mix of races and demographic backgrounds.
- Individuals are not selected separately from each racial or demographic category.

Summary
Stratified Sampling

- Divide the population directly into groups (for example, by race: Group A, Group B, Group C).
- Then take a sample from each group.

Cluster Sampling

- Choose groups that already contain a mixture of races and backgrounds.
- Then study everyone in the selected group

Hypothesis Testing

Dan had **8 boys in a row**.
If we use **p** as the probability that a child is a boy, the normal situation can be expressed as:

$$p = 0.5$$

This is called the **null hypothesis**, written as **H_0**.
Dan's claim (which he says he made before having children) was:

$$p > 0.5$$

This is called the **alternative hypothesis**, written as **H_1**.
The word **hypothesis** refers to a theory proposed either for the purpose of argument or because it is believed or suspected to be true.
An investigation of this type is usually carried out as a test called a **hypothesis test**.
The probability level at which you make the decision is called the **significance level** of the test.
Significance levels are usually expressed as percentages:

- 0.05 = 5%
- 0.01 = 1%
- etc.

Dan's Example

Hypothesis testing is a way to use data to decide if a claim is believable or if it is probably just due to chance.
We start with a "normal assumption" and then see if the evidence is strong enough to reject it.

The question could be stated as:
Test, at a 1% significance level, Dan's claim that his children are more likely to be boys than girls.
The answer would be:

Step 1: The normal assumption (Null hypothesis)

Normally, boys and girls are equally likely.
So the probability of having a boy is:

This is the **null hypothesis**:

Meaning:

"There is nothing unusual happening. Dan is just experiencing normal random chance."

Step 2: Dan's claim (Alternative hypothesis)

Dan says:

"I think I have more boys than would normally happen."

So he believes:

This is the **alternative hypothesis**:

Meaning:

"Maybe boys are actually more likely in Dan's family."

Step 3: Collect evidence

Dan has:

8 boys in a row.

The question is:

"If boys and girls are truly 50/50, how unusual is it to get 8 boys in a row?"

The probability is:

$$\left(\frac{1}{2}\right)^8 = 0.0039 = 0.39\%$$

So:

If nothing unusual is happening, there is only a **0.39% chance** of getting 8 boys in a row.

That is very unlikely.

Step 4: Compare with the significance level

The significance level is the cutoff we choose before testing.

This means:

"I will reject the normal assumption if the event is so rare that it has less than a 1% chance of happening."

Now compare:

- Probability of 8 boys: **0.39%**
- Significance level: **1%**

Since:

$0.39\% < 1\%$

the result is too unlikely under the assumption that boys and girls are equally likely.

Step 5: Make a decision

Because the evidence is very unlikely under :
we reject the null hypothesis and accept the alternative hypothesis.
Meaning:

"The data gives evidence that Dan might be right that boys are more likely."

Important: We do NOT prove Dan is right

A hypothesis test does **not** say:
"Dan is definitely correct."
It says:
"The evidence is strong enough that the normal explanation (50/50 chance) seems unlikely."

Chi-Squared Goodness-of-Fit Test

The **chi-squared test** is used to check whether observed data matches an expected distribution.
In other words, it tests whether the differences between what you observe and what you expect are simply due to chance or are statistically significant.
Definitions:

- **O = Observed value** (what you measured)
- **E = Expected value** (what you should have obtained if the data followed the expected distribution)
- **c = Degrees of freedom = number of categories − 1**

$$\chi_c^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Steps:

1. Choose a significance level **α**:
    - 1%, 5%, or 10% are the most common.
2. Perform the chi-squared calculation using your data → you obtain a **p-value**.
3. Compare the p-value with the significance level **α**:

- If:

$$p < \alpha$$

→ The differences are too large to be explained by chance → **reject the null hypothesis**.

- If:

$$p \geq \alpha$$

→ The differences could be explained by chance → **do not reject the null hypothesis**.

Example: Testing Whether a Die Is Fair

- You test whether a die is balanced.
- After 60 rolls, the chi-squared calculation gives:

$$p = 0.002$$

- Chosen significance level:

$$\alpha = 0.05\ (5\%)$$

Since:

$$0.002 < 0.05$$

we **reject the null hypothesis**. (die is balanced)

Interpretation:

There is strong evidence that the die is **not balanced**.

If you were testing whether data fits another distribution, a very small p-value (such as 0.002) means that the data **does not match the expected distribution well**.

Further Statistics not covered in the document:

Read about:

1. Poisson Distribution
2. Binomial Distribution
3. Geometric Distribution
4. Linear Combination of Random Variables
5. Monte-Carlo simulation